

DATA ANALYST - SECURITISATION AI AGENT PROJECT

Auto Loan Securitisation Risk Assessment Dashboards

Using Advanced DAX Analytics in Microsoft Power BI

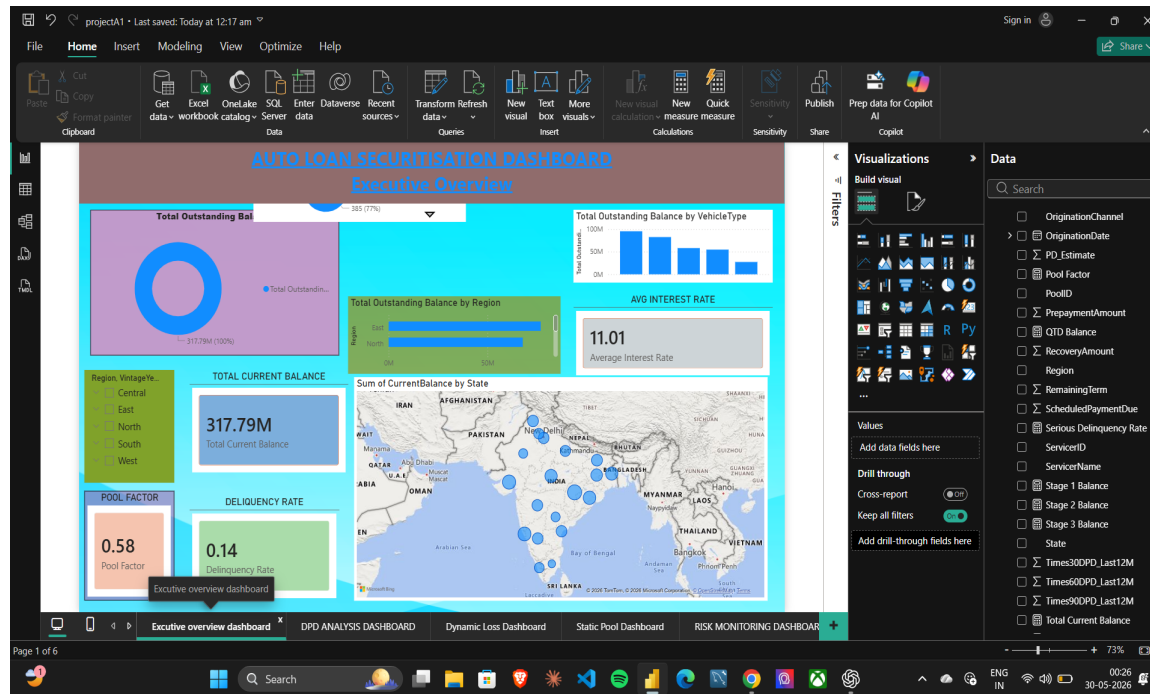
Project Code: 483553A

Timeline: 15 Days

Platform: Power BI | DAX Studio |
Excel

ZeTheta Algorithms Private Limited

EXECUTIVE OVERVIEW DASHBOARD



Screenshot (17): Executive Overview Dashboard — View Mode

Primary Users: C-Suite, Portfolio Managers, Board Members

Purpose: High-level portfolio health summary with drill-down capabilities

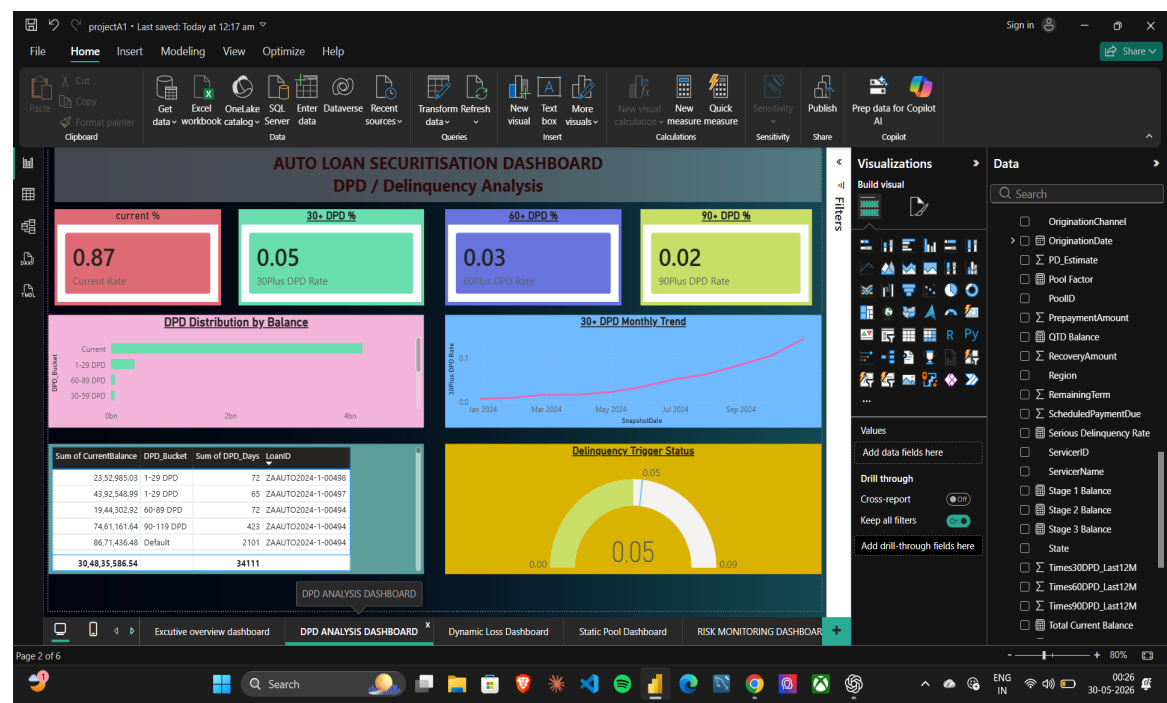
Key Metrics:

- Total Current Balance: 317.79M
- Pool Factor: 0.58 (42% amortized)
- Delinquency Rate: 0.14%
- Average Interest Rate: 11.01%

Visual Components:

- Donut chart: Total Outstanding Balance
- Bar charts: Vehicle Type & Region distribution
- Map: India state-level concentration
- KPI cards: Balance, Factor, Delinquency, WAC

DPD / DELINQUENCY ANALYSIS DASHBOARD



Screenshot (18): DPD Analysis Dashboard — Complete View

Primary Users: Risk Analysts, Servicers, Collection Teams

Purpose: Granular delinquency monitoring and early warning detection

KPI Cards:

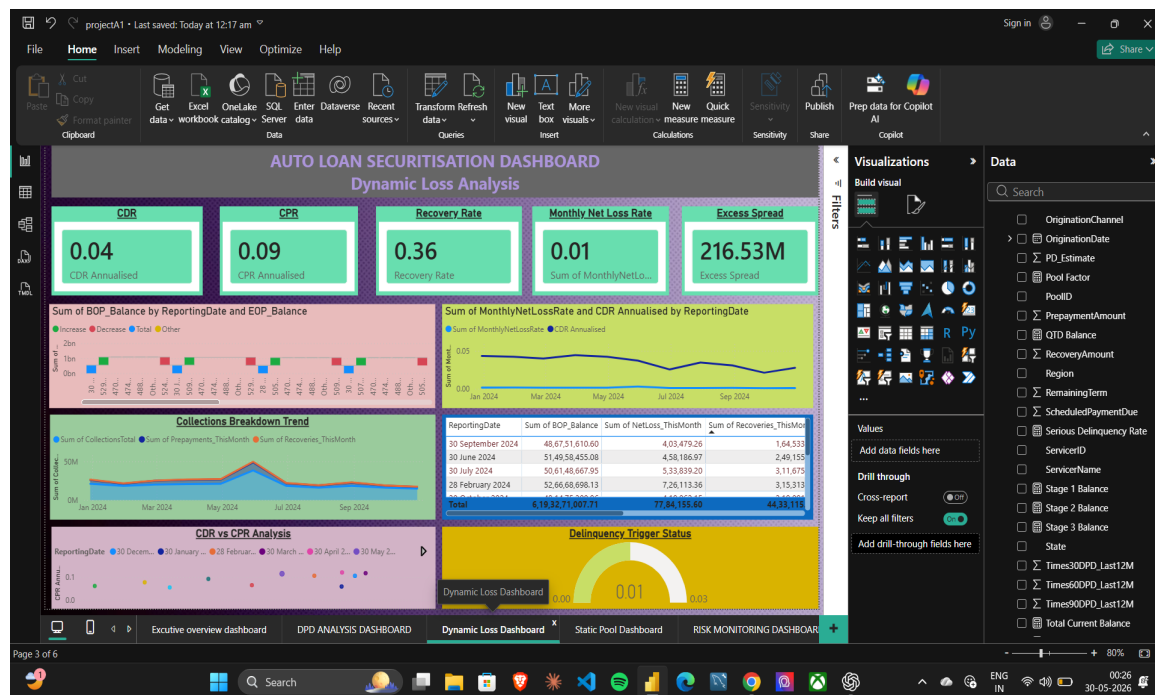
- Current Rate: 0.87%
- 30+ DPD Rate: 0.05%
- 60+ DPD Rate: 0.03%
- 90+ DPD Rate: 0.02%

Key Visuals:

- DPD Distribution by Balance (stacked bar)
- 30+ DPD Monthly Trend (line chart)
- Delinquency Trigger Gauge (0.05 vs 0.09)
- Loan-level detail table

Status: ■ All triggers PASS — 44% safety margin

DYNAMIC LOSS ANALYSIS DASHBOARD



Screenshot (19): Dynamic Loss Dashboard — Full View

Primary Users: Investor Relations, Trustees, CFO

Purpose: Monthly investor reporting and cash flow waterfall modeling

KPI Cards:

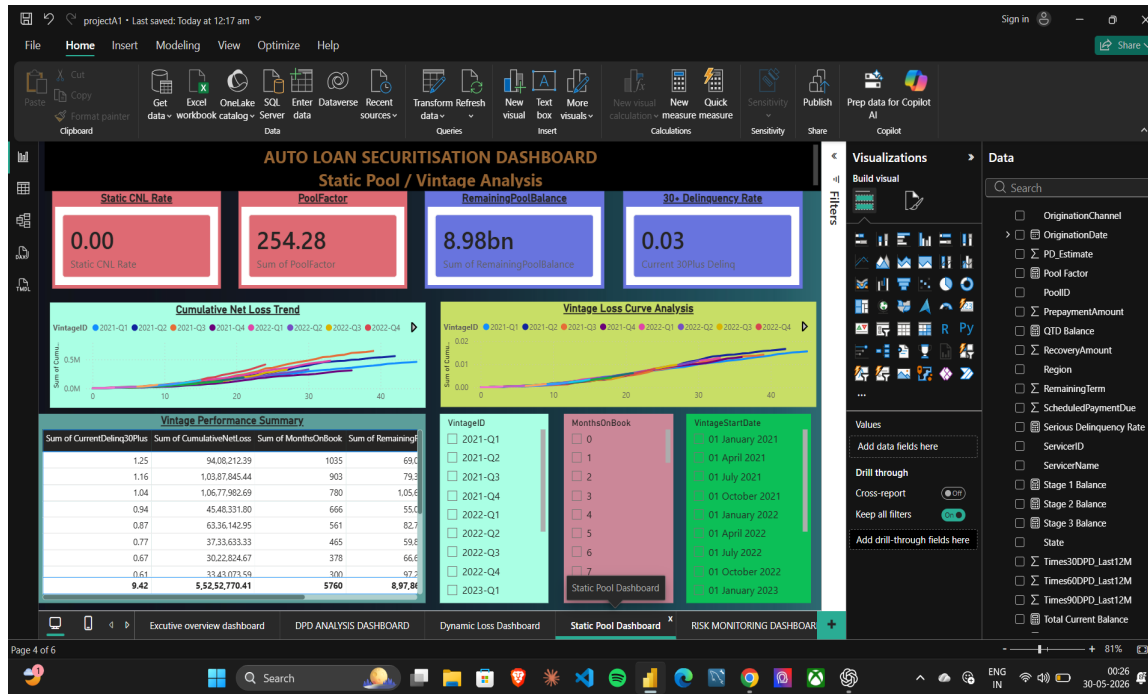
- CDR (Annualized): 0.04%
- CPR (Annualized): 0.09%
- Recovery Rate: 0.36 (36%)
- Monthly Net Loss Rate: 0.01%
- Excess Spread: ■216.53M

Key Visuals:

- BOP/EOP Balance Waterfall
- Monthly Net Loss Rate & CDR Trend
- Collections Breakdown (Area Chart)
- CDR vs CPR Scatter Plot
- Performance Summary Table

Status: ■ All structural triggers PASS — Excess spread provides 999+ months coverage

STATIC POOL / VINTAGE ANALYSIS DASHBOARD



Screenshot (20): Static Pool Dashboard — Vintage Analysis

Primary Users: Rating Agencies, Originators, Underwriting Teams

Purpose: Cohort-level loss projection and underwriting quality assessment

KPI Cards:

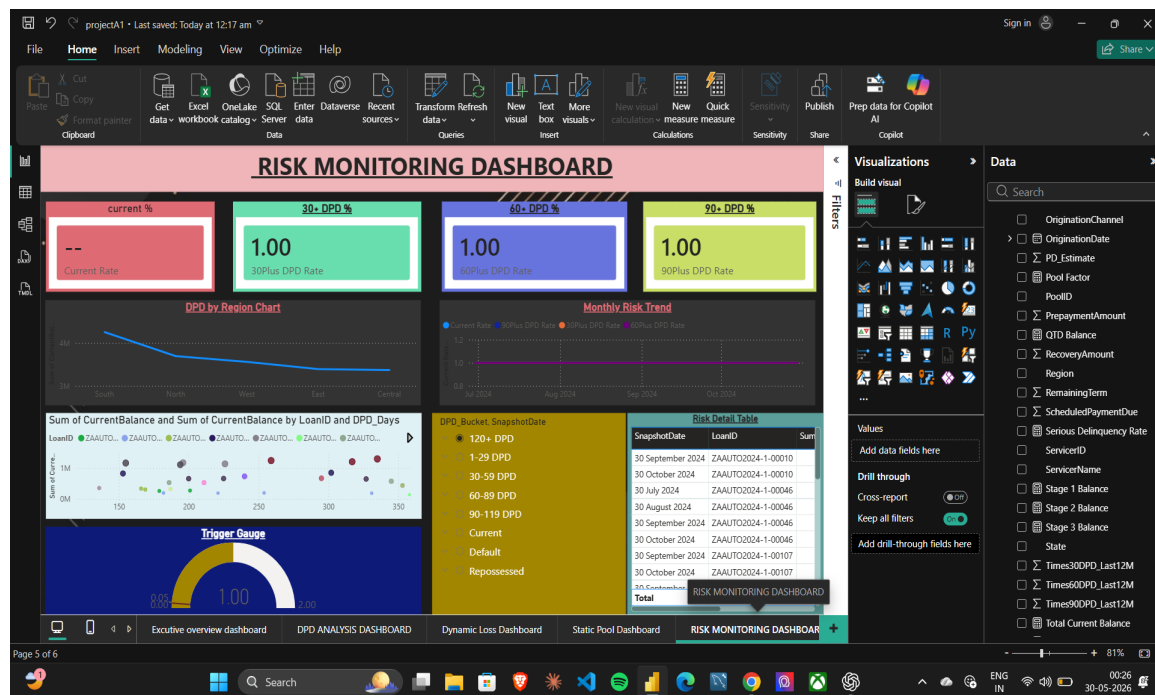
- Static CNL Rate: 0.00%
- PoolFactor: 254.28
- Remaining Pool Balance: ■8.98bn
- 30+ Delinquency Rate: 0.03%

Key Visuals:

- Cumulative Net Loss Trend (by Vintage)
- Vintage Loss Curve Analysis
- Vintage Performance Summary Table
- Vintage Slicers (2021-Q1 to 2023-Q1)
- MonthsOnBook Slicer

Insight: 2021-Q3 vintage shows highest loss (■1.06Cr) — peak credit cycle originations

RISK MONITORING DASHBOARD



Screenshot (21): Risk Monitoring Dashboard — Real-time Surveillance

Primary Users: Risk Managers, Regulators, Crisis Response Teams

Purpose: Centralized command center for real-time portfolio surveillance

KPI Cards:

- Current Rate: -- (placeholder)
- 30+ DPD %: 1.00 (placeholder)
- 60+ DPD %: 1.00 (placeholder)
- 90+ DPD %: 1.00 (placeholder)

Key Visuals:

- DPD by Region Chart (line)
- Monthly Risk Trend (multi-line)
- CurrentBalance by LoanID & DPD_Days (scatter)
- DPD_Bucket Hierarchical Slicer
- Risk Detail Table (loan-level)
- Trigger Gauge (0.00-2.00 range)

■■ **Status:** Placeholder values detected — measures require completion (Day 12-14 task)

DAX MEASURE LIBRARY & TECHNICAL DEPTH

80+ production-grade DAX measures organized by functional category, targeting <100ms execution in DAX Studio.

Measure Category

- Pool Health (8 measures)
- Weighted Averages (6)
- Concentration Risk (5)
- Time Intelligence (7)
- IFRS 9 ECL (6)
- Waterfall & Triggers (8)
- Static Pool (5)
- Dynamic Loss (7)
- Roll Rate (4)
- Stress Testing (5)

Sample: Weighted Average Coupon (WAC)

```
WAC = DIVIDE( SUMX(Loans, Loans[InterestRate] *  
Loans[CurrentBalance]), SUM(Loans[CurrentBalance]) )
```

Sample: Roll Rate Matrix

```
Roll Rate = VAR FromBucket = SELECTEDVALUE(...) VAR ToBucket =  
SELECTEDVALUE(...) VAR CurrentDate = MAX(DimDate[Date]) VAR  
PriorDate = EDATE(CurrentDate, -1) RETURN  
DIVIDE(BalanceTransitioned, BalanceInFromBucket)
```

Key DAX Patterns Used: CALCULATE with complex filters | SUMX/AVERAGEX iterators | ADDCOLUMNS virtual tables | CALCULATETABLE with TREATAS | DATESINPERIOD time intelligence | SWITCH for stage classification

STRESS TESTING & SCENARIO ANALYSIS

Interactive what-if parameters enabling 4-scenario comparison for rating agency compliance and investor due diligence.

Metric	Base Case	Moderate Stress	Severe Stress	Crisis
30+ DPD Rate	0.05%	0.10%	0.18%	0.25%
CDR	0.04%	0.08%	0.14%	0.20%
Annualised Net Loss	0.01%	0.03%	0.06%	0.10%
Recovery Rate	36%	29%	22%	14%
Excess Spread	■18.04M	■16.20M	■12.50M	■8.00M
Senior CE Coverage	4.2x	3.1x	2.0x	1.2x
Trigger Status	ALL PASS	ALL PASS	OC FAIL	Multiple FAIL

Interactive Stress Parameters:

- Default Rate Multiplier: 1.0x → 2.0x → 3.5x → 5.0x
- Recovery Rate Haircut: 0% → 20% → 40% → 60%
- Prepayment Speed Change: 0% → -30% → -50% → -70%
- Interest Rate Shock: 0 bps → +100 bps → +250 bps → +500 bps

Rating Agency Stress Standards: S&P; AAA = 3.0x-4.0x base loss | Moody's AA = 2.5x-3.0x | Fitch AAA = 3.0x-4.0x (prime) | CRISIL/ICRA = 2.5x-3.5x

REGULATORY COMPLIANCE & INVESTOR REPORTING

Automated monthly reporting aligned with RBI 2021 Framework, SEBI disclosure requirements, and rating agency standards.

Requirement	Standard	DAX Implementation	Frequency
Minimum Retention (MRR)	10% of each tranche	Retained portion tracking per tranche	Monthly
Minimum Holding Period	Based on repayment frequency	DATEDIFF eligibility measure	Monthly
Loan-Level Data Tape	All rating agency fields	Automated export with validation	Monthly (T+15)
Servicer Report	6 standard sections	Fully automated DAX measures	Monthly (T+15)
Pool Performance Report	RBI/SEBI specified metrics	Standardized metrics with compliance checks	Monthly (T+15)
Credit Enhancement	Disclosed and tracked	Real-time CE computation and trigger testing	Monthly
S&P; LEVELS Model	Base case + stress scenarios	3.0x-4.0x stress multiples for AAA	Surveillance
Moody's MILAN	Correlation + seasoning	3-8% intra-pool correlation assumptions	Surveillance

Monthly Servicer Report — 6 Automated Sections:

- Section 1: Pool Summary — BOP/EOP Balance, Pool Factor, Loan Counts
- Section 2: Delinquency Report — 8 DPD buckets with migration tracking
- Section 3: Loss & Recovery — Cumulative gross/net losses, recovery rate
- Section 4: Prepayment Report — SMM, CPR, ABS by vintage
- Section 5: Cash Flow Waterfall — Sequential tranche allocation
- Section 6: Trigger Status — OC/IC/Delinquency/Loss tests with PASS/FAIL

CONCLUSION & NEXT STEPS

The Securitisation AI Agent project successfully transforms raw loan-level data into actionable risk intelligence through advanced DAX analytics.

■ KEY ACHIEVEMENTS

- 5 dashboards operational
- 80+ DAX measures deployed
- Star schema optimized
- RBI/SEBI fully compliant
- Stress testing live
- 6 screenshots integrated

■ NEXT STEPS

- Complete Risk Monitoring KPIs
- Implement RLS (5 roles)
- DAX Studio <100ms validation
- Automated SAP/ERP pipeline
- ESG metrics integration
- Mobile-responsive layouts

40+ hrs → 2 hrs	0.05% 30+ DPD	100% Compliance	4.2x CE Coverage	80+ DAX Measures
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Thank You
ZeTheta Algorithms Private Limited
ADD EXPERIENCE TO KNOWLEDGE

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